

OSSP WEEK 2 PRACTICAL REPORT

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Task 1: Process Creation using fork(), exec(), and wait()

Objective: Develop a C program that accepts a Linux command, creates a child process using fork(), executes the command using exec(), waits for the child process using wait(), and displays the Parent and Child Process IDs.

Figure 1: Writing the C program in Nano Editor



speed@DESKTOP-P4G181S: ~



GNU nano 8.7.1

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main() {
    char command[100];
    pid_t pid;

    printf("Enter Linux command: ");
    scanf("%s", command);

    pid = fork();

    if(pid < 0) {
        printf("Fork failed!\n");
        return 1;
    }

    if(pid == 0) {
        printf("Child Process\n");
        printf("Child PID: %d\n", getpid());

        execlp(command, command, NULL);

        printf("Invalid command\n");
    }
    else {
        printf("Parent Process\n");
        printf("Parent PID: %d\n", getpid());
        printf("Child PID: %d\n", pid);

        wait(NULL);

        printf("Child process completed.\n");
    }

    return 0;
}
```

Figure 2: Compiling the program and executing the 'ls' command

```
speed@DESKTOP-P4G181S: ~ X + v
wsl: Unknown key 'wsl2.autoMemoryReclaim' in C:\Users\speed\.wslconfig:3
speed@DESKTOP-P4G181S:~$ nano process.c

speed@DESKTOP-P4G181S:~$ gcc process.c -o process
speed@DESKTOP-P4G181S:~$ ./process
Enter Linux command: ls
Parent Process
Parent PID: 464
Child PID: 465
Child Process
Child PID: 465
Hello JOE auto-execute.md hi.c process process.c projects
Child process completed.
speed@DESKTOP-P4G181S:~$ |
```

Figure 3: Executing the 'pwd' command

```
speed@DESKTOP-P4G181S:~$ ./process
Enter Linux command: pwd
Parent Process
Parent PID: 485
Child PID: 486
Child Process
Child PID: 486
/home/speed
Child process completed.
speed@DESKTOP-P4G181S:~$ |
```

Task 2: Hardware Resources and Operating System Services

The following Linux commands were executed to study operating system services and hardware abstraction:•
uname -a• lscpu• ps• top

Figure 4: Output of lscpu, ps and top

```
speed@DESKTOP-P4G181S: ~  
NUMA:  
  NUMA node(s): 1  
  NUMA node0 CPU(s): 0-7  
Vulnerabilities:  
  Gather data sampling: Not affected  
  Ghostwrite: Not affected  
  Indirect target selection: Not affected  
  Itlb multihit: Not affected  
  L1tf: Not affected  
  Mds: Not affected  
  Meltdown: Not affected  
  Mmio stale data: Not affected  
  Old microcode: Not affected  
  Reg file data sampling: Not affected  
  Retbleed: Mitigation; Enhanced IBRS  
  Spec rstack overflow: Not affected  
  Spec store bypass: Mitigation; Speculative Store Bypass disabled via prctl  
  Spectre v1: Mitigation; usercopy/swapgs barriers and __user pointer sanitization  
  Spectre v2: Mitigation; Enhanced / Automatic IBRS; IBPB conditional; PBRSE-eIBRS SW sequence; BHI BHI_DIS_S  
  Srbds: Not affected  
  Tsa: Not affected  
  Tsx async abort: Not affected  
  Vmscape: Not affected  
speed@DESKTOP-P4G181S:~$ ps  
  PID TTY          TIME CMD  
  321 pts/0    00:00:00 bash  
  520 pts/0    00:00:00 ps  
speed@DESKTOP-P4G181S:~$ top  
top - 06:39:33 up 6 min, 1 user, load average: 0.03, 0.02, 0.00  
Tasks: 25 total, 1 running, 24 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 1.2 us, 0.4 sy, 0.0 ni, 97.3 id, 1.1 wa, 0.0 hi, 0.1 si, 0.0 st  
MiB Mem : 15773.6 total, 14102.5 free, 789.5 used, 1079.0 buff/cache  
MiB Swap: 4096.0 total, 4096.0 free, 0.0 used, 14984.1 avail Mem  


| PID | USER     | PR | NI | VIRT  | RES   | SHR   | S | %CPU | %MEM | TIME+   | COMMAND         |
|-----|----------|----|----|-------|-------|-------|---|------|------|---------|-----------------|
| 46  | root     | 19 | -1 | 50396 | 16756 | 15464 | S | 0.3  | 0.1  | 0:00.10 | systemd-journal |
| 1   | root     | 20 | 0  | 24132 | 15424 | 11624 | S | 0.0  | 0.1  | 0:00.39 | systemd         |
| 2   | root     | 20 | 0  | 3180  | 2208  | 2072  | S | 0.0  | 0.0  | 0:00.00 | init-systemd(Ub |
| 6   | root     | 20 | 0  | 3180  | 2044  | 1940  | S | 0.0  | 0.0  | 0:00.00 | init            |
| 78  | systemd+ | 20 | 0  | 22416 | 14508 | 12012 | S | 0.0  | 0.1  | 0:00.03 | systemd-resolve |
| 86  | root     | 20 | 0  | 35468 | 12356 | 9152  | S | 0.0  | 0.1  | 0:00.21 | systemd-udev    |


```

Observation

CPU: The operating system schedules processor time among processes.
Memory: The OS allocates and manages memory for each process.
Storage: The OS manages files and directories through the file system.
I/O Devices: Device drivers allow applications to communicate with hardware.

Conclusion

The practical successfully demonstrated Linux process creation using `fork()`, command execution using `exec()`, synchronization using `wait()`, and the abstraction of hardware resources through operating system services.